Economics 5040: Dynamic Stochastic Modeling
Lecture 3: Choice Under Uncertainty and Expected Utility

1 From Dated Commodities to Uncertain Commodities

In the first week, we treated consumption today and consumption tomorrow as different commodities. A unit of consumption delivered today is not the same commodity as a unit delivered next year. The gross real interest rate R describes the market tradeoff between these dated commodities.

We now introduce uncertainty. Suppose that we know today's consumption, but tomorrow one of several possible *states of the world* will occur. For example, tomorrow's income might be high or low. Consumption delivered tomorrow in the high-income state is now a different commodity from consumption delivered tomorrow in the low-income state.

Thus a commodity can be distinguished not only by its physical characteristics and its date of delivery, but also by the state of the world in which it is delivered.

Let the set of possible states tomorrow be

$$\{1, \ldots, S\},$$

and let

$$\pi_s > 0, \qquad \sum_{s=1}^{S} \pi_s = 1$$

denote their probabilities.

A state-contingent consumption plan is

$$(c_1(1), c_1(2), \ldots, c_1(S)).$$

Before we study markets in which people trade claims to these state-contingent commodities, we need a theory of how people rank uncertain prospects.

2 Lotteries

A *lottery* is a probability distribution over possible (real money) outcomes.

Suppose there are two possible consumption outcomes:

$$c_H > c_L.$$

A simple lottery can be written

$$L = (p, c_H; 1 - p, c_L),$$

meaning that the consumer receives c_H with probability p and c_L with probability $1-p$.

More generally, if the possible outcomes are

$$x_1, \ldots, x_n,$$

a lottery is a vector

$$L = (p_1, \ldots, p_n), \qquad p_i \geq 0, \qquad \sum_{i=1}^{n} p_i = 1,$$

where p_i is the probability of outcome x_i.

We assume that people have preferences over lotteries. Thus

$$L \succeq M$$

means that lottery L is regarded as at least as good as lottery M.

The question is whether these preferences can be represented in a particularly convenient form.

3 The von Neumann–Morgenstern Axioms

As before, we begin with *completeness* and *transitivity*.

Completeness says that any two lotteries can be compared. Transitivity says that these comparisons are internally consistent.

To obtain expected utility, we add two important assumptions.

Continuity

Suppose

$$L \succ M \succ N.$$

Continuity says that there exists some probability $p \in (0,1)$ such that

$$M \sim pL + (1-p)N.$$

In words, if M lies between a better lottery L and a worse lottery N, then there is some mixture of L and N that is just as good as M.

This assumption rules out preferences in which an arbitrarily small change in probabilities causes a discontinuous jump in the ranking of lotteries.

Independence

The independence axiom says that if

$$L \succeq M,$$

then, for any other lottery N and any $0 < p < 1$,

$$pL + (1-p)N \succeq pM + (1-p)N.$$

The same lottery N has been added to both alternatives with the same probability. Independence says that doing so should not reverse the original ranking between L and M.

This is the crucial additional restriction that gives expected utility its special linearity in probabilities.

4 The von Neumann–Morgenstern Expected Utility Theorem

We now give a simple version of the representation theorem for a finite set of outcomes.

Suppose the possible outcomes can be ranked

$$x_1 \prec x_2 \prec \cdots \prec x_n,$$

where x_1 is the worst outcome and x_n is the best outcome.

Theorem 1 (von Neumann–Morgenstern)**.** *If preferences over lotteries satisfy completeness, transitivity, continuity, and independence, then there are numbers $u(x_1), \ldots, u(x_n)$ such that any lottery*

$$L = (p_1, \ldots, p_n)$$

can be represented by

$$U(L) = \sum_{i=1}^{n} p_i u(x_i).$$

Thus lotteries are ranked according to expected utility.

A Constructive Proof

The proof contains the main economic idea, so it is worth understanding.

First normalize the utility of the worst and best outcomes:

$$u(x_1) = 0, \qquad u(x_n) = 1.$$

We can do this normalization because an monotonic transformations of a utility functions represents the same underlying preferences. We learned this fact last week.

Now consider any intermediate outcome x_i. Since

$$x_n \succ x_i \succ x_1,$$

continuity implies that there is some probability q_i such that

$$x_i \sim q_i x_n + (1 - q_i) x_1.$$

Define

$$u(x_i) = q_i.$$

This gives a very concrete interpretation to the utility number $u(x_i)$: it is the probability of receiving the best outcome in a best–worst lottery that makes the consumer indifferent to receiving x_i for certain. Notice that $q_1 = 0$ and $q_n = 1$.

Now consider *any* arbitrary lottery

$$L = (p_1, \ldots, p_n).$$

Whenever outcome x_i occurs, replace it by the best–worst lottery to which the consumer is indifferent:

$$x_i \sim u(x_i) x_n + [1 - u(x_i)] x_1.$$

By the independence axiom, making these substitutions does not change the consumer's ranking of the original lottery L. After reducing the resulting compound lottery, the probability of receiving the best outcome x_n is

$$\sum_{i=1}^{n} p_i u(x_i) = \sum_{i=1}^{n} p_i q_i$$

The probability of receiving the worst outcome is therefore

$$1 - \sum_{i=1}^{n} p_i u(x_i) = 1 - \sum_{i=1}^{n} p_i q_i.$$

Hence

$$L \sim \left(\sum_{i=1}^{n} p_i u(x_i) \right) x_n + \left(1 - \sum_{i=1}^{n} p_i u(x_i) \right) x_1.$$

So any lottery can be reduced to one that considers only the best and worst outcomes. Indeed, continuity has allowed us to reduce every outcome to a lottery between the best and worst outcomes. Independence then guarantees that, among these special lotteries, more probability on the best outcome is better. Thus a lottery between the best and worst outcomes is preferred when it gives a higher probability to the best outcome. Therefore the original lottery L is ranked by

$$\boxed{U(L) = \sum_{i=1}^{n} p_i u(x_i).}$$

That is expected utility.

5 Expected Utility and Ordinary Utility Are Different

There is an important contrast with the utility functions introduced in Lecture 1.

For ordinary choice under certainty, if $u(x)$ represents preferences, then any strictly increasing transformation

$$v(x) = g(u(x))$$

represents the same preferences.

Expected utility is more restrictive.

Suppose

$$U(L) = \sum_{i} p_i u(x_i).$$

If we replace u by

$$\widetilde{u}(x) = a + bu(x), \qquad b > 0,$$

then

$$\sum_{i} p_i \widetilde{u}(x_i) = a + b \sum_{i} p_i u(x_i),$$

which preserves the ranking of lotteries.

Thus a von Neumann–Morgenstern utility function is unique up to a *positive affine transformation*:

$$\boxed{\widetilde{u} = a + bu, \qquad b > 0.}$$

An arbitrary increasing transformation will generally not preserve expected utility rankings.

6 Consumption Risk

We now apply expected utility to consumption.

Suppose tomorrow's consumption is random. In state s, consumption is $c_1(s)$, and the state occurs with probability π_s.

Expected utility from tomorrow's consumption is

$$E[u(c_1)] = \sum_{s=1}^{S} \pi_s u(c_1(s)).$$

If current consumption c_0 is known and preferences are time separable, a natural two-period utility function is

$$U = u(c_0) + \beta E[u(c_1)].$$

Thus

$$\boxed{U = u(c_0) + \beta \sum_{s=1}^{S} \pi_s u(c_1(s)).}$$

This expression will be central to the remainder of the course.

Notice that two conceptually distinct assumptions are being used. Time separability tells us how utility at different dates is combined. Expected utility tells us how utility across uncertain states is combined.

7 Risk Aversion

Consider a random consumption level C with expected value

$$E[C].$$

A consumer is *risk averse* if receiving the expected consumption for certain is at least as good as facing the risky consumption prospect:

$$u(E[C]) \geq E[u(C)].$$

If u is concave,

$$u''(c) < 0,$$

then this inequality follows from Jensen's inequality:

$$\boxed{u(E[C]) \geq E[u(C)].}$$

Thus concavity of the von Neumann–Morgenstern utility function represents risk aversion.

The economics is intuitive. When marginal utility diminishes with consumption, the utility lost from an unfavorable consumption shock is larger than the utility gained from an equally large favorable shock.

8 A Simple Example

Suppose a consumer has logarithmic utility

$$u(c) = \log c$$

and faces the lottery

$$C = \begin{cases} 50 & \text{with probability } 1/2, \\ 150 & \text{with probability } 1/2. \end{cases}$$

Expected consumption is

$$E[C] = 100.$$

But expected utility is

$$E[u(C)] = \frac{1}{2}\log 50 + \frac{1}{2}\log 150.$$

Because the logarithm is strictly concave,

$$E[\log C] < \log E[C] = \log 100.$$

The consumer therefore prefers receiving 100 for certain to facing the 50–150 lottery, even though both have the same expected consumption.

9 The Certainty Equivalent

The *certainty equivalent* of a lottery is the certain amount of consumption that makes the consumer indifferent between receiving that amount and facing the lottery.

Let CE denote the certainty equivalent. It is defined by

$$u(CE) = E[u(C)].$$

Therefore

$$\boxed{CE = u^{-1}\left(E[u(C)]\right).}$$

For a risk-averse consumer,

$$CE < E[C]$$

whenever the lottery contains genuine risk.

In the logarithmic example,

$$\log CE = \frac{1}{2}\log 50 + \frac{1}{2}\log 150.$$

Hence

$$CE = \exp\left(\frac{1}{2}\log 50 + \frac{1}{2}\log 150\right) = \sqrt{50 \cdot 150}.$$

Thus

$$CE \approx 86.60.$$

The consumer is indifferent between the risky prospect with expected consumption 100 and approximately 86.60 units of consumption for certain.

10 The Risk Premium

The *risk premium* is the difference between expected consumption and the certainty equivalent:

$$\boxed{RP = E[C] - CE.}$$

In our example,

$$RP = 100 - 86.60 = 13.40.$$

The risk premium measures how much expected consumption the consumer would be willing to give up in order to eliminate the risk.

A risk-neutral consumer has

$$CE = E[C]$$

and hence a zero risk premium.

A risk-averse consumer has

$$CE < E[C]$$

and hence a positive risk premium.

11 Measuring Risk Aversion

Concavity tells us whether a von Neumann–Morgenstern utility function represents risk aversion, but we would also like to measure *how risk averse* a consumer is. Two standard measures are the coefficients of *absolute risk aversion* and *relative risk aversion*.

Suppose that $u(c)$ is twice differentiable, with

$$u'(c) > 0$$

and

$$u''(c) < 0.$$

The *coefficient of absolute risk aversion* is defined as

$$\boxed{A(c) = -\frac{u''(c)}{u'(c)}.}$$

This measure, introduced by Arrow and Pratt, compares the curvature of the utility function with its slope. Greater curvature, relative to marginal utility, corresponds to greater aversion to small risks.

Absolute risk aversion is particularly useful for thinking about risks whose size is measured in *units of consumption*. For example, we might ask how a consumer feels about a gamble in which consumption increases or decreases by \$100, regardless of the consumer's initial level of consumption.

The *coefficient of relative risk aversion* is defined as

$$\boxed{\rho(c) = -\frac{cu''(c)}{u'(c)} = cA(c).}$$

Relative risk aversion is useful when risks are naturally expressed as a *fraction* of consumption. Instead of asking how a consumer responds to gaining or losing a fixed number of units, we might ask how the consumer responds to a gamble that increases or decreases consumption by, say, ten percent.

The distinction between absolute and relative risk aversion is therefore important. Absolute risk aversion concerns the response to risks of a fixed absolute size, while relative risk aversion concerns the response to risks whose size changes in proportion to consumption.

For example, consider logarithmic utility,

$$u(c) = \log c.$$

Then

$$u'(c) = \frac{1}{c}$$

and

$$u''(c) = -\frac{1}{c^2}.$$

Hence absolute risk aversion is

$$A(c) = -\frac{-1/c^2}{1/c} = \frac{1}{c},$$

while relative risk aversion is

$$\rho(c) = cA(c) = 1.$$

Thus logarithmic utility exhibits *decreasing absolute risk aversion*: as consumption rises, the consumer becomes less averse to a risk of a fixed absolute size. At the same time, logarithmic utility exhibits *constant relative risk aversion*: the consumer's attitude toward risks that are proportional to consumption does not change with the level of consumption.

More generally, an important class of utility functions is

$$u(c) = \begin{cases} \dfrac{c^{1-\rho} - 1}{1-\rho}, & \rho \neq 1, \\ \log c, & \rho = 1. \end{cases}$$

For these preferences,

$$-\frac{cu''(c)}{u'(c)} = \rho.$$

The parameter ρ is therefore the coefficient of relative risk aversion. These preferences are called *constant relative risk aversion*, or CRRA, preferences.

We will use CRRA preferences frequently in dynamic economics. The parameter ρ will play an important role not only in attitudes toward risk but also, when the same period utility function is used in an intertemporal model, in the willingness to substitute consumption across dates.

12 Why This Matters for Dynamic Economics

Risk aversion gives consumers a reason to want insurance.

If consumption is likely to be high in one state and low in another, a risk-averse consumer would generally like to transfer consumption from the high-consumption state to the low-consumption state.

This raises a natural economic question:

> Can markets allow people to trade consumption across states of the world in the same way that borrowing and lending allow them to trade consumption across dates?

The answer leads directly to state-contingent claims and asset pricing.

In the next lecture, we will assign prices to consumption delivered in particular states of the world. We will see that an asset is fundamentally a claim to a vector of state-contingent payoffs, and that the prices of those claims depend on both probabilities and marginal utility.

This will also allow us to distinguish between *complete markets*, in which arbitrary state-contingent consumption plans can be traded, and *incomplete markets*, in which they cannot.